MASA03 Lab 3

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Introduction to Estimators

Let $X \sim F_\theta,\ \theta \in \Theta$ and $x_1, \ldots, x_n$ independent observations of X. We wish to find a candidate for what *theta* might be based on these observations. To this end, we construct what is called an *estimator*. An estimator is simply a function $h : \mathbb{R}^n \to \Theta$. There are multiple commonly used estimators. For example: plug-in estimator, maximum-likelihood estimator and least-square estimator.

1 Confidence Intervals

Suppose that $x_1, \ldots, x_n$ are independent observations of $X \sim F_\theta$, where $\theta \in \Theta$ is a parameter on which the distribution function depends. Let $\hat{\theta} = h(x_1, \ldots, x_n)$ be an estimator of θ (e.g. the plug-in estimator). Then we can view $\hat{\theta}$ as a random variable by setting $\hat{\theta} = h(X_1, \ldots, X_n)$ where $X_k \sim F_\theta$. Then, if $I \subseteq \Theta$ is such that $\mathbb{P}(\hat{\theta} \in I) = 1 - \alpha$, we say that I is a confidence region with level $1 - \alpha$ (if $\Theta \subseteq \mathbb{R}$ and I is an interval, it is called a confidence interval).

2 Hypothesis Testing

Suppose that $x_1, \ldots, x_n$ are independent observations of $X \sim F_\varphi$ where $\varphi \in \Theta$. Form hypotheses

$$\begin{cases} H_0 : \varphi \in A \\ H_1 : \varphi \notin A \end{cases},$$

statistic $T : \mathbb{R}^n \to \mathbb{R}$, *critical region* $C \subseteq \mathbb{R}$, and reject our hypothesis H_0 if $T(x_1, \ldots, x_n) \in C$. In order to define T and C define the power function

$$\pi(\theta) = \mathbb{P}\left(T\left(X_\theta^{(1)}, \ldots, X_\theta^{(n)}\right) \in C\right)$$

where $X_\theta^{(i)} \sim F_\theta$. We want to choose the function T and the critical region C such that $\alpha \coloneqq \sup_{\theta \in A} \pi(\theta)$ is small and for any $\theta \notin A$ we have $\pi(\theta) > \alpha$. This construction will give that if our data was distributed as F_θ with $\theta \in A$, it is unlikely that our hypothesis is rejected, and if $\theta \notin A$ it is more likely that our hypothesis is rejected. Ideally we want $\pi(\theta)$ to be much bigger than α for $\theta \notin A$. In the case that $A = \{\theta_0\}$ (i.e. A is a singleton), the corresponding H_0 is then called a *simple* null hypothesis, and the supremum in the definition of α is just $\pi(\theta_0)$.

3 Exercises

3.1 1.

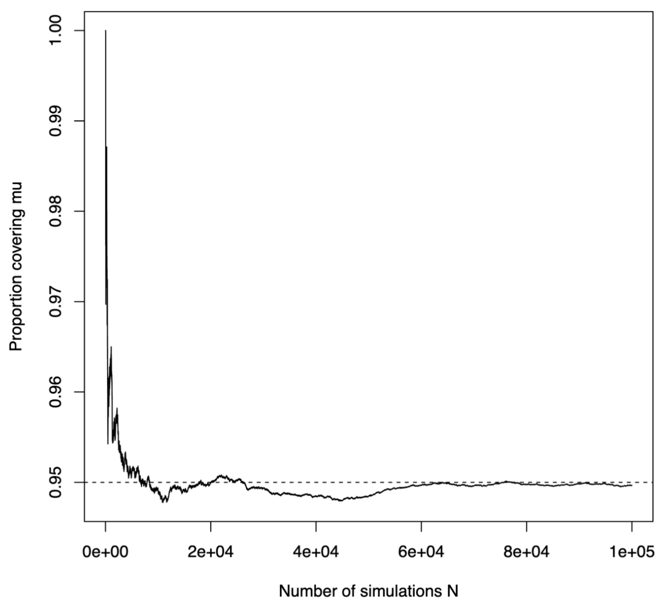

Figure 1: Exercise 1

3.1.1 (a)

The simulated covering proportion is close to 0.95, and the running coverage plot stabilizes near 0.95, so the theoretical value appears correct.

3.1.2 (b)

$$I_i = \mathbf{1}\{\mu \text{ lies inside the } i\text{th simulated confidence interval}\},$$

The variables $I_1, \ldots, I_N$ are i.i.d. Bernoulli(0.95). By the Law of Large Numbers,

$$\frac{1}{N}\sum_{i=1}^{N} I_i \longrightarrow 0.95 \quad \text{as } N \to \infty.$$

Hence the measured proportion of intervals containing μ converges to the true coverage probability 0.95.

3.2 2.

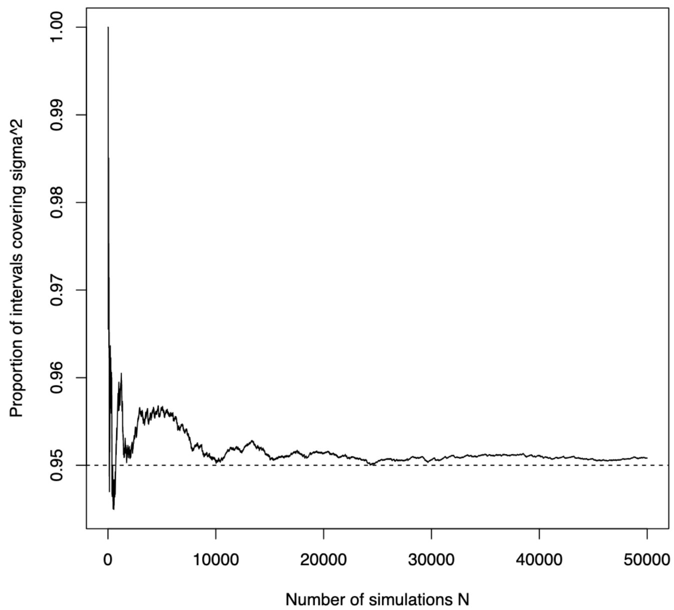

Figure 2: Exercise 2

3.2.1 (a)

The simulated covering proportion is close to 0.95, and the running coverage plot stabilizes near 0.95, so the theoretical value appears correct.

3.2.2 (b)

$$I_i = \mathbf{1}\{\sigma^2 \text{ lies inside the } i\text{th simulated confidence interval}\}.$$

The variables $I_1, \ldots, I_N$ are i.i.d. Bernoulli with mean 0.95. By the Law of Large Numbers,

$$\frac{1}{N}\sum_{i=1}^{N} I_i \longrightarrow 0.95 \quad \text{as } N \to \infty.$$

Hence the empirical proportion of intervals covering σ^2 converges to the true coverage probability 0.95.

3.3 3.

3.3.1 (a)

The simulated rejection proportion is close to 0.05, so the test attains the intended significance level.

3.3.2 (b)

Let I_i denote the indicator that the ith simulated test rejects H_0. The variables $I_1, \dots, I_N$ are i.i.d. Bernoulli with mean 0.05. By the Law of Large Numbers,

$$\frac{1}{N}\sum_{i=1}^{N} I_i \longrightarrow 0.05 \quad \text{as } N \to \infty.$$

Thus the empirical rejection proportion converges to the true significance level.

3.4 4

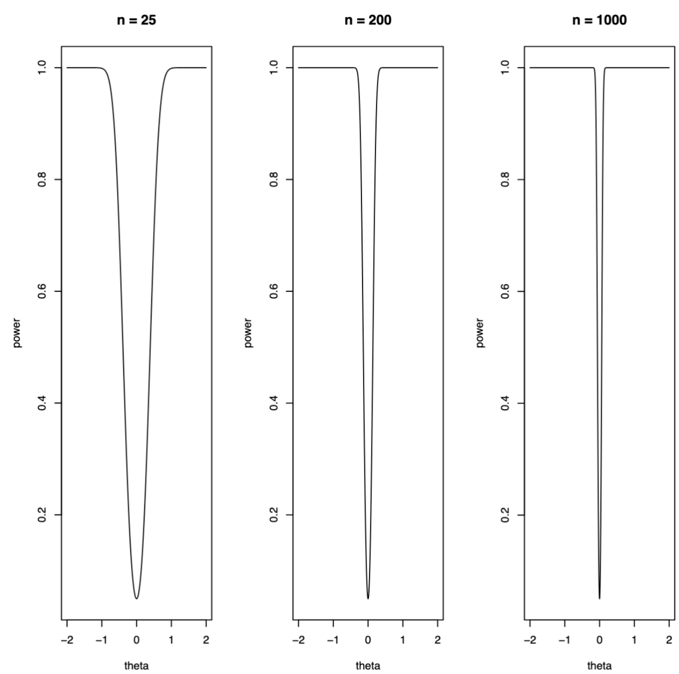

Figure 3: Enter Caption

3.4.1 (a)

When $n = 200$, the power curve has the expected shape: it stays near 0.05 at $\theta = 0$, and then rises as θ moves away from 0. The test rarely rejects when the null is true, and it becomes increasingly likely to reject when the true mean differs from 0.

3.4.2 (b)

With only $n = 25$ points, the power curve is much flatter. The test has noticeably less ability to detect nonzero values of θ. In other words, smaller samples make it harder for the test to distinguish between H_0 and H_1.

3.4.3 (c)

For $n = 1000$, the power curve becomes very steep. Even small changes in θ away from 0 lead to power close to 1. This happens because with a very large sample, the test can detect even tiny deviations from the null hypothesis.